

Embodied Telepresent Connection (ETC): Exploring Virtual Social Touch Through Pseudohaptics

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ABSTRACT

In search of missing tactile sensations in telepresent virtual reality (VR), we set off to explore how we might be able to apply pseudo-haptics to elicit an embodied illusion of social touch. We developed ETC (Embodied Telepresent Connection), a prototype that allows two participants to interact remotely through pseudo-haptic touch via their abstract aura avatars. ETC consists of a set of interaction patterns aimed to elicit an illusion of interpersonal touch using embodied metaphors, simulated physics, integration of visuals and sounds, biosignal representation, and embedded social connotations. We showed ETC at 5 events and 3 workshops, observing participants and gathering feedback from them. Here, we report on our design process and initial observations relating to the experience of virtual social touch. We observed that pseudo-haptics applied to social interaction afford a subtle and somewhat uncanny experience of virtual touch with the potential for intimate embodied connection.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Interaction design theory, concepts and paradigms**.

KEYWORDS

social touch, virtual reality, pseudo-haptics, embodiment, social connection

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1 INTRODUCTION

Touch is fundamental to how we experience the physical world and connect with others [7, 14, 39]. Touch simultaneously defines the extent of the body and provides rich information about the world that surrounds it. According to philosopher Mark Hansen, the typical vision-centred approach to Virtual Reality (VR) disrupts the characteristic interplay between touch and the other senses [17]. Yet, touch is important for feeling embodied in a virtual body [26, 35]. In VR, touch enhances spatial presence and “gives us the strongest impression of being directly in contact with external objects” [31, p.172]. Touch also facilitates social presence (the sense of being with another) [33] and contributes to social development [7] and connection [14]. Despite its importance, social touch has seen limited investigation in Human-Computer Interaction (HCI) [53], and VR [13]. Because of our increasing reliance on technology that does not afford touch for social interactions, our need for social touch has increased. Field refers to this phenomenon as “touch hunger.” [12] The lack of social touch limits our capacity to appropriately satisfy our need for touch and our ability to communicate socially through touch.

In the growing field of Social VR, interpersonal touch is often overlooked and excluded from taxonomies [25] and designs for non-verbal communication [50]. Consumer experiences that do incorporate touch often do so via vibrotactile interfaces (e.g., B-Haptics suit in VR Chat). These simply stimulate the sense of touch rather than reproducing anywhere near its full complexity and richness [13]. While there are several devices seeking to provide richer experiences of touch (e.g., those discussed in [37]), these devices are often bulky and expensive. As an alternative approach, pseudo-haptics utilize visual and auditory stimuli to evoke tactile illusions [8]. Pseudo-haptics may be a promising research direction as it

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has already been demonstrated to improve sensations of touch and resistance over haptic vibration motors [42].

Studies investigating social touch within VR often focus on collaboratively manipulating an object rather than interpersonal touch [33]. To address this gap in research on the effects of social touch in VR we set out to explore *how we might elicit experiences of social touch between immersants using pseudo-haptics and a popular consumer VR system*. We contribute our design along with preliminary findings from a series of workshops and demonstrations.

2 RELATED WORK

2.1 Designing for connection and mediated social touch

Social connection is crucial for our health and happiness [29, 41], which motivates a large body of work in HCI exploring how technology can facilitate and stimulate the experience of connection [18, 30, 34]. Stepanova et al.'s review of systems for genuine connection across art and design identifies mediated social touch as a key strategy for supporting genuine connection [47]. Mediated touch can support intimacy in long-distance close relationships [45] and encourage new connections with strangers [19]. While the majority of mediated touch research explores some haptic interface [16], recent research shows that for social touch, visuals alone may also perform no worse than current haptic technologies [56]. In social VR applications, the lack of tactility is often addressed by haptic hardware such as gloves [21]. Some co-located VR experiences, often in the field of VR theatre or dance, also may utilize actors' bodies touching the immersant to provide a tactile experience [2, 55]. *MachineToBeAnother* [9] and *Body RemiXer* [11] use this strategy to create an interpersonal embodied interaction using the participants' own bodies. However, haptic interfaces and physical co-presence are not always feasible, necessitating other solutions to support tactility in social VR.

2.2 Pseudo-haptics in virtual reality

Pseudo-haptics occur as a result of multi-modal integration of sensory stimuli where the visual and auditory senses can evoke tactile sensations [38]. Biocca et al. found even simple visual representations of spring-like resistance could evoke physical sensations of force [5]. The field of pseudo-haptics has seen growing interest in both screen-based interaction and immersive VR [32]. Studies have also demonstrated the potential of audio-haptic illusions [8]. While pseudo-haptics can improve sensations of touch and resistance along with increased immersion and enjoyment [42], vibrotactile haptics continue to dominate VR. Despite being the subject of research since at least 2001 [5], pseudo-haptics has seen little application for social touch. One study by Sykownik et al. [48] applied pseudo-haptics to social touch in VR. This study did not explore the tactile experience itself but focused on comparing emotional responses to high-intimacy contact (e.g., hugging) versus low-intimacy contact (e.g., high-fiving). Unlike Sykownik et al.'s realistic visuals, we chose to use abstract representations that avoid the uncanny valley while providing a canvas for exploring a wide array of pseudo-haptics beyond the everyday. We also used hand-tracking to allow for direct interaction and physical simulation of the fingers reacting to virtual objects.

Combining pseudo-haptics' promise to evoke touch with social connection research, we investigate pseudo-haptics for social interaction. We performed an exploratory study, approved by the Simon Fraser University Research Ethics Board, integrating an iterative design process with a qualitative study to uncover immersants' virtual social touch experiences.

3 SYSTEM DESIGN

Price et al. suggest that "looking beyond realism as a target for design allows for the integration of creative artistic approaches to touch design, framed by the affordances of technology." [37] Accordingly, for Embodied Telepresent Connection (ETC) we developed a series of experimental visual and auditory interactions that might evoke an experience of social touch via pseudo-haptics on common VR hardware. A video of the ETC prototype is available on YouTube¹ and the program can be downloaded from our website². The video is essential to fully understand the dynamic effects in ETC that can be difficult to convey through images alone.

The design process followed an intuitive and artistic approach where "the creative process forms the pathway through which new insights, understandings and products come into being." [6, p.46]. It builds on knowledge established in *Body RemiXer* [11], another VR experience centered around social touch and other social connection research [47]. We began by creating a series of interactive "sketches" that explored the limits of human representation including aura-like energy fields, fluid hands that could merge with other bodies, and dissolving bodies forming a galaxy [10]. From those experiments and referring to research into illusions of touch and physiological communication [8, 38, 42], we identified the key elements which we incorporated into the final prototype presented here. Those elements were selected based on reflections on our experiences with the sketches given hardware constraints and their fit within the aims of the research. We included visual and auditory responses to proximity, touch, pressure, and heartbeat.

Rather than aim for photorealism we created abstract, particle-based avatars to facilitate possibilities for manipulating the avatar similar to how Bernal et al. expressed emotions in particle-based avatars [3]. Even simple abstract avatars can evoke social presence [4] and pseudo-haptic effects [5]. The abstract avatar consists of confetti-like particles continuously generated across the surface of the body (Figure 1). They remain on the surface of the body and typically consist of cooler teal and purple colours. A cloud-like heart hovers inside the avatar's chest. The pseudo-haptic elements are similarly abstract, focusing on auditory and visual interactions that fit the overarching aesthetic.

3.1 Implementation

Implemented in the Unity game engine [52], the prototype can connect two people remotely using a Meta Quest 2. It uses Photon PUN2 [36] to network hand and head tracking data, and heartrate between two immersants. RootMotion's VRIK [44] is used to position the avatar's bodies via inverse kinematics and Hand Physics Toolkit [15]. The abstract particle bodies are then generated locally using Unity's VFX Graph. When no one else is present, immersants

¹ETC Prototype Video: youtu.be/3OW05odYNHI

²APK available for Meta Quest 1/2/Pro: ispace.iat.sfu.ca/project/etc/

can interact with a clone avatar that mirrors their movements, allowing for solo exploration. Each pseudo-haptic element can be selectively enabled or disabled allowing for different combinations.

3.2 Proximity interactions

While pseudo-haptics research typically focuses on touch and resistance to movement, we also incorporated interaction modalities that respond to proximity. This recognizes the virtual bodies' capacities to interact with one another and the proxemic extension of social presence into surrounding space.

Visual Warmth. This parameter makes the particles warmer in color as the immersants approach each other aligning the metaphorical link between hue and heat [57] connoting emotional warmth [27, 54]. From about 1.5m away the particles begin subtly becoming redder until, when touching, they become entirely red and orange near the area of contact. We chose to use this familiar metaphor in an effort to evoke a sense of warmth from the other avatar, overcoming the typically cold, lifeless appearance.

Particle Attraction. When immersants move within approximately 10 cm of each other their particles are progressively attracted towards each other as they come closer to touching. We incorporated this element to encourage immersants to move towards touching and in response to early observations among our team that this kind of particle flow could produce very subtle sensations akin to a warm tingle.

Tremolo Flute. The auditory response to proximity is a medium-pitched wavering flute-like sound. This sound gets louder as immersants get closer aligning with the building tension of getting closer together in a kind of harmonic dissonance. As identified by Benford et al., carefully integrated experiences of discomfort can be critical to producing meaningful experiences and facilitate social bonding [1]. In line with their model, this tremolo flute sound produces a kind of rising action towards the climax of making physical contact. It also acknowledges the uncanny strangeness of approaching another telepresent person, simultaneously feeling their building presence and absence.

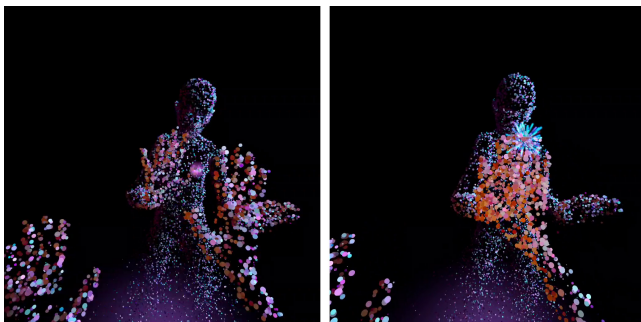


Figure 1: Proximity and tactile interactions. Left: the hands subtly become redder as they become closer together. Right: on contact, the hands are red and a small firework is emitted from the point of contact

3.3 Contact interactions

We incorporated interactions acknowledging moments of interpersonal touch. These interactions were meant to signify a positive moment and respond well to common forms of social touch such as holding hands and high-fiving. Contact is detected using physics colliders and works anywhere on the body. The visual representation is a small firework emitted from the location of touch. While originally a placeholder, participants in an early workshop enjoyed this fun and rewarding experience akin to a spark lighting between them. The auditory representation of touch is a randomly pitched 'ping' elicited with each point of contact. This created a playful interaction that aligned with the whimsical fireworks and encouraged immersants to explore. Such playfulness can stimulate social connection [47], akin to work by Hoby [19].

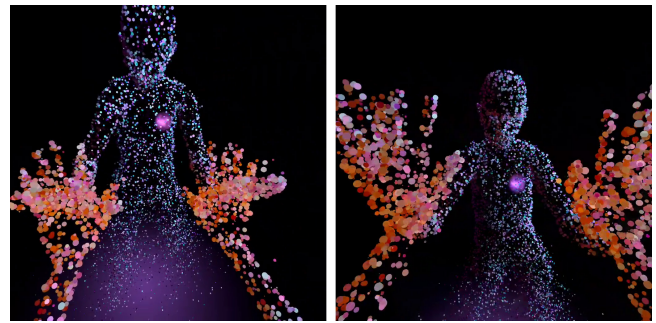


Figure 2: Left: with pseudo-haptics disabled the hands pass through each other with no resistance. Right: enabled, the hands push against each other, displacing proportionally to the force—as seen here this begins to push the hands upwards and outwards since equal force is being applied by both immersants

3.4 Pseudo-haptic resistance

Perhaps the most commonly researched aspect of pseudo-haptics is resistance to an exerted force evoked through visual displacement [8]. As seen in Figure 2, by turning this pattern off, immersants' avatars would directly follow their actual position passing through each other's bodies. Turning it on activates a physics simulation that displaces the displayed position of the avatar, preventing bodies from passing through each other.

Physics Simulation. In ETC, we implemented this resistance using a combination of VRIK [44] for inverse kinematics based on hand and head tracking, and HPTK [15] to simulate the physics body. This incorporates physics colliders to simulate the effect of touching through a second 'puppet' body. The immersant's real position is tracked using an invisible avatar while a second avatar is physically simulated and is pulled towards the actual position. The force exerted is proportional to the displacement between the two. For example, if one immersant holds their hand in place and the other pushed through it, they would see their hands displace to roughly halfway in-between. This was designed to allow for a range

of pseudo-haptic interactions from fine interactions between intertwining fingers, to performing complex handshakes and feeling resistance when pushing against each other's chest.

Audio force-feedback. The audio component of resistance was a low hum that grew louder with increasing force. The spring stretching sounds used in other implementations would not seem to fit the ethereal appearance of our avatars. Instead, the sound is more akin to the sound of electrical arcs, fitting with the aura-like energy field appearance of the avatars.

3.5 Intimate heartbeat

Representing biosignals, such as heartbeat, on VR avatars can foster social connection and intimacy [3, 24, 28], because of its connotation for life and closeness [22]. To evoke a sense of intimate touch, when immersants touch each other near the heart they hear the other's heartbeat (Figure 3). A realistic heartbeat sound stands out from the other more synthetic and abstract audio to emphasize an intimate experience. The heartbeat can be subtly heard as the hand gets closer, slowly getting louder before reaching a maximum with altered tonality similar to using the arm as a stethoscope when contact is made.

The heartbeat is visualized through a cloud-like heart that oscillates in size and colour with each beat. The body particles also become very subtly redder in a wave that begins from the heart and spreads outward. While ETC was designed to communicate real-time heart rate using a Movesense or Apple Watch, this feature was not used for the workshops given our focus on pseudo-haptics. Instead the heart rate varied between 60 and 80 bpm to evoke a relaxed yet realistically variable heartbeat.

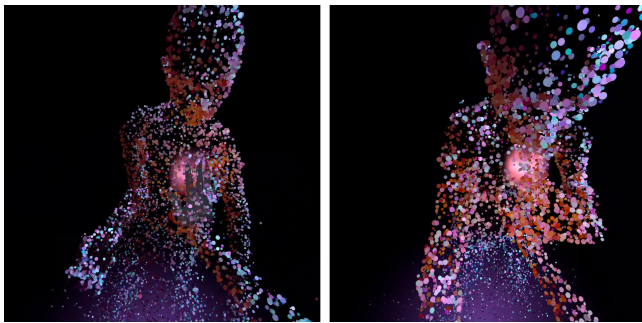


Figure 3: Touching the heart as seen here allows the immersant to hear the other person's heartbeat. Left: the position of the hand is highlighted by a dark grey outline for clarity. Right: the other immersant pulls the participant's hand in closer while simultaneously feeling their heartbeat.

4 INITIAL EVALUATION

4.1 Demos and workshops

We demonstrated ETC at two conferences, three lab visits, and conducted three somaesthetics-inspired workshops [20] to help participants attune and articulate their bodily sensations. We observed over 50 immersants combining observational methods [43], phenomenological interviews [23], and somatic reflections [51].

The workshops were typically conducted in separate rooms to give participants privacy, except for conference demonstrations that were performed in a busy conference environment. Pairs of participants interacted with each other whenever possible, sometimes while in the same physical space, and sometimes deliberately placed in different rooms to prevent verbal communication. When only one participant was present, the researcher would take the role of the other participant. Most demonstrations involved showing participants the prototype with all features on. Later somaesthetics-inspired workshops began by allowing participants to interact with an unknown other participant before trying different combinations of prototype elements together. At this early stage of research, the diversity of set-ups allowed us to explore how the context might shape participants' experience and what settings might offer most fruitful avenues for further investigation. The researchers made an effort to avoid priming participants with any specific expectations about the experience, and thus didn't provide any instructions about possible interaction or who they will be interacting with, instead allowing participants to discover it for themselves.

Following their experience, we conducted individual or group interviews where we prompted participants with open questions, inviting to describe their experience and any sensations they had. A comprehensive data analysis is ongoing. Here we report selected observations related to the experience of touch using pseudonyms for privacy.

4.2 Observations

4.2.1 Faint Bodily Sensations. When prompted to describe their experience of touch, participants would often note that they could feel some faint ineffable sensations on their body, for example in their hands or shoulders. For instance, Marion described feeling the **warmth** when seeing the colour change to orange and red. While Sandra was even surprised to report how her hands felt sweaty as if she was actually shaking someone's hand. Denise felt similar warmth from just the pseudo-haptic resistance, without the colour change. Others described a perplexing, indescribable sensation of **pressure or resistance** when experiencing the pseudohaptic resistance and not being able to go through others' hands [Matt]. Some participants described a **tingling** sensation on their skin when seeing the fireworks as the particles pop and explode from their hand [Maya]. Some participants also experienced a certain sense of **texture**, e.g., Mia described a "crackling" sensation of her hair being moved around on her skull from the "fireworks" when Mark was rubbing her head: *"the sound effects were happening on my side, it was cool, and weird, because the sounds you hear when your head is rubbed is normally like that [shows by moving her hair around], but instead I was hearing this crackling."* She continued, *"It made me feel like I was more in this virtual body, because the reflection of the physical effect on me, the sounds that were being made when I was being touched were different than the ones in real life."* After Mark initiated the rubbing, Mia leaned in and was surprised that *"it felt almost as nice as having your head rubbed in real life."* Maria described the **uncanny quality** of the sensations: *"it was a bit more than conceptual. But at the same time I knew it wasn't physical touch. So it was something else ... there is this moment where it's hard to know if something is real or not."*

4.2.2 Pseudo-haptic resistance. Resistance was a subtle parameter that participants typically did not notice until they tried it both on and off, or we drew their attention to it. This aligns with Pusch and Lécuyer’s finding that priming can enhance pseudo-haptics [38] especially when such experiences are atypical in VR. Some participants found that it felt “sticky” [Dan], others described a feeling of “resistance” [Matt], while others said that the other body felt more “solid and real” [Diane]. Yet, this realness was not always desired. Resistance made immersants feel like their own and others’ bodies were more **solid, and grounded in reality**. However, several immersants observed that this may not support the feeling of interpersonal connection that ETC was designed for [Denise]. Instead, they emphasized that VR affords the opportunity to have impossible experiences and new captivating ways to connect. E.g., by merging bodies when resistance was disabled, immersants found the experience **mesmerizing and poetic** [Drew]. Even with resistance on, some participants would seek this experience of merging, by pushing through and stepping inside other’s body. This way, Mila enjoyed blending her body with the other. She found it “magical” and felt “dizzying, but in a good way [...] it felt very intimate, but not in the way when I physically touch someone.” For some participants the **real world physics** of pseudo-haptics **trivialized the experience**. Others wanted a more subtle effect that better aligned with the ethereal aesthetics [Dan]—bodies that only partly resisted force, more like entering a force field as in [38].

4.2.3 Compounding visuals and sounds. Participants often described the critical role of sound in the bodily experience of the moment of touch. Visuals and sounds seem to combine to create a **single perceptual event** in line with the multi-sensory integration theory underlying pseudo-haptics. Mark observed that “some effects can substitute haptics [...] It was a bit more satisfying because there was sound.” However, Mark also noted how this evoked a desire for more tactility: “At times like this I wish that haptic suits were more common place, because I expected to feel pressure. And it did threw me off[...] when you just go through the person.” Mila noted how the sounds made touch **game-like**: “When the hands touched each other I felt like I was winning points, I was achieving something.” For Maria the integration of sound visual brought **sense of realness**: “it feels somehow a little bit more real, when there was the sound and the image change because of that. It gave like another dimension somehow.” She then elaborated that because of this enhanced realness of touch she realized she is actually interacting with Maya, rather than a virtual image.

4.2.4 Touching hearts. Seeing the heart affected the perception of the other avatar, it “made it more human, more intimate, more close” [Maria]. The heart attracted immersants’ attention, **inviting touch**. Mia describes that “because the heart was the biggest thing [...] I wanted to see if I can interact with it [makes a poking gesture].” Mila also felt that the gradient of light comprising the heart was the most attractive thing to follow to interact. Mila continued: “I am kinda missing the sense of touch, but I think the visual experience is strong enough that I could somewhat feel the heartbeats in my hands, when I put my hand around it [...] I felt the **physical presence that was subtle** but was enough for me to perceive it as a heartbeat, or holding someone’s heart in between my hands.” She elaborated how she liked touching the heart, and didn’t get a direct response.

Here the sound just grew stronger and stronger as she got closer and closer, which made this interaction more meaningful for her, without cheapening it through a game-like mechanic of reward for interaction. “I want to put it as elegantly as I felt... It was really subtle, really caring, it was like holding something immature, that can’t really respond to me, but I can feel explicitly that I care for it. It almost seemed like it was **representative of the life form**.” Mila described the feeling of the heart: “I could feel a small tingling effect in my hands, it was not like the actual sensation of the heartbeat, it was like... there were multiple little vibrations [...] it was like how you would feel compressed air.”

4.2.5 Powerful experience. ETC elicited fairly strong reactions in many participants. For some, it was profoundly positive—a beautiful experience of spiritual **connection**. Mila described the experience as “very strong and intimate.” While Maria felt it was “**nearly uncanny, strange, but cool**.” Especially one of the earlier prototypes, where the abstract avatar had fairly defined facial features, felt sometimes creepy and unsettling [Maya]. Maria said: “It’s strange to describe, I don’t have many words for it. It was strange, but nice? in a way.” Mark also described the experience as “surreal [...] it was kinda an alien encounter experience,” while feeling that it has a strong potential for an **intimate “almost romantic”** experience, that he did not feel open to experiencing as he was aware of the surrounding professional context.

5 DISCUSSION

Most research on VR touch focuses on simulating physics, usually focused on grabbing objects. Yet, for social touch, besides physical properties, there is also a crucial component of the social and emotional connotations associated with interpersonal touch [37]. EEG research shows that we begin to experience sensations of social touch even prior to contact through anticipatory excitation [40]. In ETC, everything from distant proximity-based interactions to intimately touching hearts amplified the sensations experienced. While touch enhanced the realization of the realness of the human behind the avatar (e.g., Maria’s realization of her interaction with Maya), in turn, the social connotation of being touched imbued the feeling of touch with meaning and amplified sensations given past experiences (e.g., Sandra’s sweating hands). Our workshop findings demonstrate that beyond pseudo-haptic force feedback, there are many ways to evoke embodied illusions that arise through aligned audiovisual elements, metaphors, and social connotations. While many experiments isolate a single sense, congruous audiovisual stimuli were typically preferred in our workshops.

We found that pseudo-haptic sensations can be subtle and unlike ordinary touch, but most participants indicated feeling ‘something’ (e.g., temperature or tingling). This is significant, as any research on pseudo-haptics should recognize that the phenomenology of pseudo-haptic experiences may not directly align with experiences of ordinary physical touch. Pseudo-haptic research requires careful and open-minded approaches that encourage immersants to notice and reflect upon its subtleties (e.g., somaesthetics [20]). Immersants may have felt something strange and unique, yet tactile, that they might not describe as ‘touch.’

Our participants had meaningful social experiences that sometimes verged on the profound. In fact, ethereal bodies might help to

align visual representations with such subtle and strange sensations. While most VR research uses realistic avatars (e.g., [48]), the visual physicality of these avatars likely does not align with pseudo-haptic sensations as they afford an expectation of physical experiences of touch. As Sykownik and Maic identify, VR offers an opportunity to go beyond the physical and augment tactile interaction in new ways [49]. We suggest that more than just augmenting realistic avatars this can be directly ingrained into the design of the avatars themselves.

Reciprocally, the pseudo-haptic effects need to be designed to align with the unique visual affordances of such an abstract appearance. The concrete touch was notably misaligned with the ethereal representation and some participants felt that such rigid pseudo-haptic resistance could interfere with experiences aiming for social connection. Yet, others enjoyed the resistance, reflecting the multiplicity of experiences that can emerge from social touch. Pseudo-haptics are not a one-dimensional catch-all design pattern for every experience but a tool that requires careful alignment with the unique desired experience. Used in this way pseudo-haptics can be used to bring the body and tactile sense into even the most ethereal experience.

5.1 Future directions

Future research should investigate the role of both the physical simulation and the social aspect of interpersonal touch in VR. We plan to delve further into the combination of biosignals with pseudo-haptics, extending upon the heartbeat interaction. Breathing, heartbeat, electromyography (EMG), electroencephalograph (EEG), are all promising biosensors that offer opportunities to create even richer social experiences that can bring avatars to life and reveal social and emotional cues across distances.

There are also many more interaction patterns that could be explored and this particular abstract particle body is just one possibility among many. Future work should accordingly investigate how such abstract representations compare to realistic ones such as [48] and the effect on pseudo-haptic sensations and social presence. Our design process led us to use abstraction to avoid the uncanny valley and acknowledge the others' lack of physicality, however, the effect of such design decisions have yet to be directly compared.

While we focused here on the exclusive use of pseudo-haptics, a promising application is combining them with haptic interfaces which has already resulted in positive outcomes with vibrohaptic controllers [42]. Integrating haptic gloves or other devices may allow for even richer tactile experiences.

Another intriguing avenue of research that opens up with the development of technology for telepresent social touch in VR, is its potential to explore touch across different agents, both artificial and human. Similar to the self-dialogue approach to VR therapy [46], the 'clone' mode in ETC might offer an opportunity to explore self-compassion through bodily interaction, by allowing immersants to "get in touch" with themselves.

6 CONCLUSION

While this work is still in early stages, the ETC prototype unveiled the potential of pseudo-haptic social touch and inspiring further

research. Taking an artistic approach we developed creative pseudo-haptic elements that embraced technological constraints while transcending the everyday physical experience. Our contribution is both a demonstration of the intriguing possibilities and a window into the subtle nuances of pseudo-haptics. We were able to elicit tactile sensations through *simulated physics, ethereal visuals and sounds, embodied metaphors, and embedded social connotations*, extending previous findings that abstract [5] and extraordinary stimuli such as force fields [38] may in fact be best for evoking pseudo-haptic sensations.

Through a combination of creative, qualitative, and quantitative approaches we can advance towards a better understanding of the potential to elicit social tactility and embodied connection through pseudo-haptics, leading to a richer multi-sensory future for social VR where touch is not left behind.

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